Sneha Patra **D20A** **43**

Blockchain Experiment 3

AIM: Create a Cryptocurrency using Python and perform mining in the Blockchain created.

THEORY:

Q1: Challenges in P2P networks.

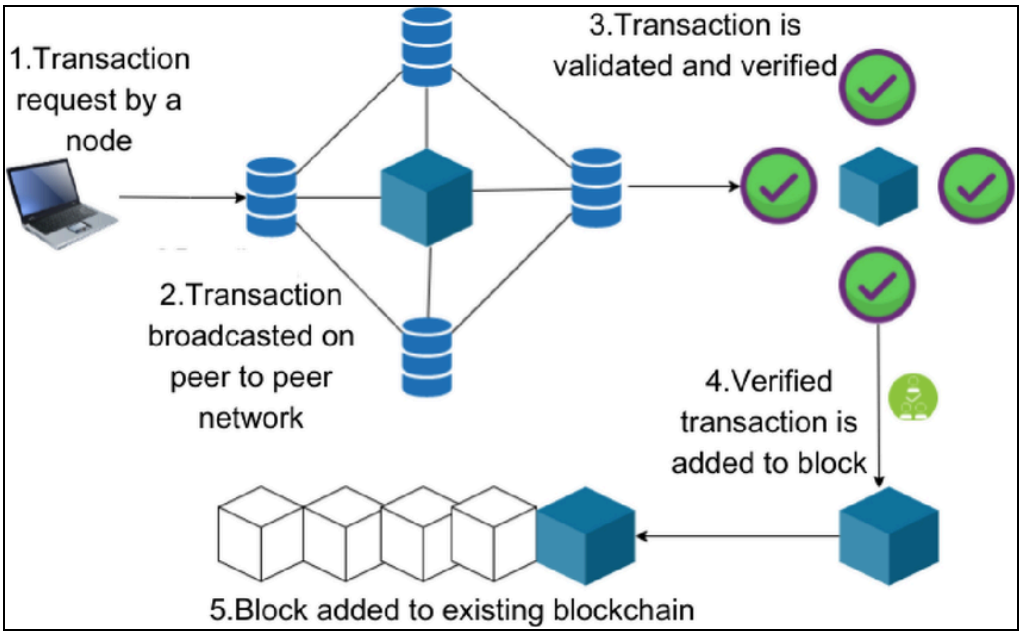
Peer-to-Peer (P2P) networks are decentralized systems where nodes communicate directly without a central authority. Although they provide transparency and fault tolerance, several challenges exist:

- **Security Issues**
  Nodes may behave maliciously and attempt attacks such as double-spending or fake transactions.
  Sybil attacks occur when one entity creates multiple fake identities to control the network.
- **Network Latency**
  Since nodes are distributed globally, data propagation can be slow.
  Delays in block propagation may lead to temporary forks in the blockchain.
- **Consensus Management**
  All nodes must agree on a single version of the blockchain.
  Maintaining consensus requires complex algorithms like Proof-of-Work or Proof-of-Stake.
- **Scalability**
  As more nodes join, communication overhead increases.
  High transaction volume can slow down processing.
- **Resource Consumption**
  Mining consumes high computational power and electricity.
  Storage requirements grow as the blockchain size increases.

Q2: How transactions are performed on the network?

Blockchain transactions follow a structured process:

- **Step 1: Transaction Creation**
  A user creates a transaction containing sender address, receiver address, and amount.
  The transaction is digitally signed using the sender's private key.
- **Step 2: Transaction Broadcast**
  The transaction is broadcasted to all nodes in the P2P network.
- **Step 3: Verification**
  Nodes verify Digital signatures, Account balance, Transaction validity
- **Step 4: Mempool Storage**
  Valid transactions are stored temporarily in the mempool awaiting confirmation.
- **Step 5: Mining and Block Creation**
  Miners select transactions from the mempool.
  Transactions are grouped into a block and mined using Proof-of-Work.
- **Step 6: Block Addition**
  The new block is added to the blockchain after consensus.
  Transactions become permanent and immutable.



Q3: Explain the role of mempools

A mempool (memory pool) is a temporary storage area for unconfirmed transactions.
Functions of Mempools:

- Stores pending transactions before they are added to a block.
- Helps miners select transactions for block creation.
- Prevents network congestion by organizing transaction queues.

Q4: Write briefly about the libraries and the tools used during implementation.

Libraries used:

- **datetime**
  Used to generate timestamps for each block created in the blockchain.
- **hashlib**
  Used to perform SHA-256 hashing for block identification and proof-of-work mining.
- **json**
  Used for encoding and decoding blockchain data into JSON format for API communication.
- **uuid4 (from uuid module)**
  Used to generate a unique identifier for each node in the blockchain network.
- **urlparse (from urllib.parse)**
  Used to extract and manage network node addresses.
- **Flask**
  Used to create RESTful API endpoints such as mine_block, get_chain, add_transaction, connect_node, and replace_chain for blockchain operations.
- **request (from Flask)**
  request is used to receive input data from API calls.
- **jsonify (from Flask)**
  jsonify is used to return blockchain responses in JSON format.
- **requests**

Used for communication between different blockchain nodes to implement consensus and longest chain replacement.

Tools used:

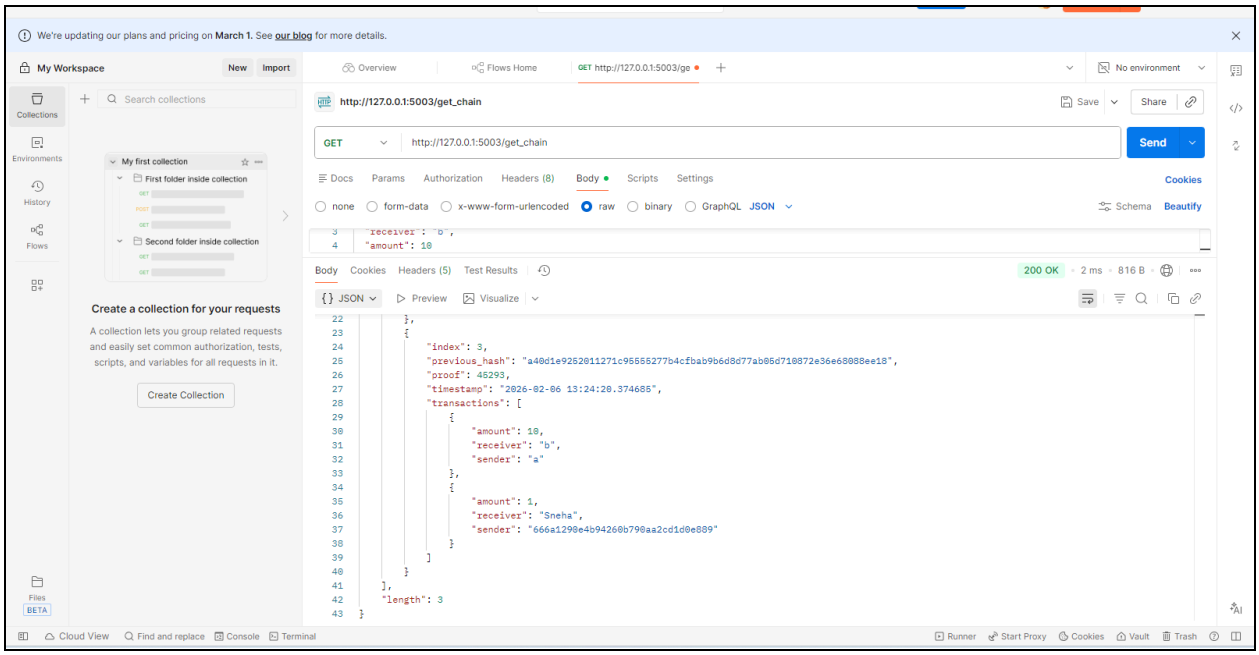
- **VS Code**
  Used as the development environment to write and execute Python code.
- **Thunder Client (VS Code Extension)**
  Used to test REST API endpoints such as mining blocks, adding transactions, connecting nodes, and applying consensus.

Output:

1. Download hadcoin_node_5001.py, hadcoin_node_5002.py, hadcoin_node_5003.py.
2. Install required packages.
3. Run each node in separate terminals.
4. Connect nodes using Postman – POST `/connect_node`.


```
{
  "nodes":
  [
    "http://127.0.0.1:5002",
    "http://127.0.0.1:5002"
  ]
}
```


5. Add transactions – POST
6. Add 3 transactions in 5001
7. Condition 2 — Each node must have **different chain lengths**


We're updating our plans and pricing on March 1. See our blog for more details.
My Workspace
New
Import
Overview
Flows Home
GET http://127.0.0.1:5003/ge
No environment
Collections
Environments
History
Flows
Search collections
My first collection
First folder inside collection
Second folder inside collection
Create a collection for your requests
A collection lets you group related requests and easily set common authorization, tests, scripts, and variables for all requests in it.
Create Collection
http://127.0.0.1:5003/get_chain
Save
Share
GET
http://127.0.0.1:5003/get_chain
Send
Docs
Params
Authorization
Headers (8)
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Scripts
Settings
Cookies
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form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Schema
Beautify
"receiver": "b",
"amount": 10
Body
Cookies
Headers (5)
Test Results
200 OK
2 ms
816 B
JSON
Preview
Visualize
},
{
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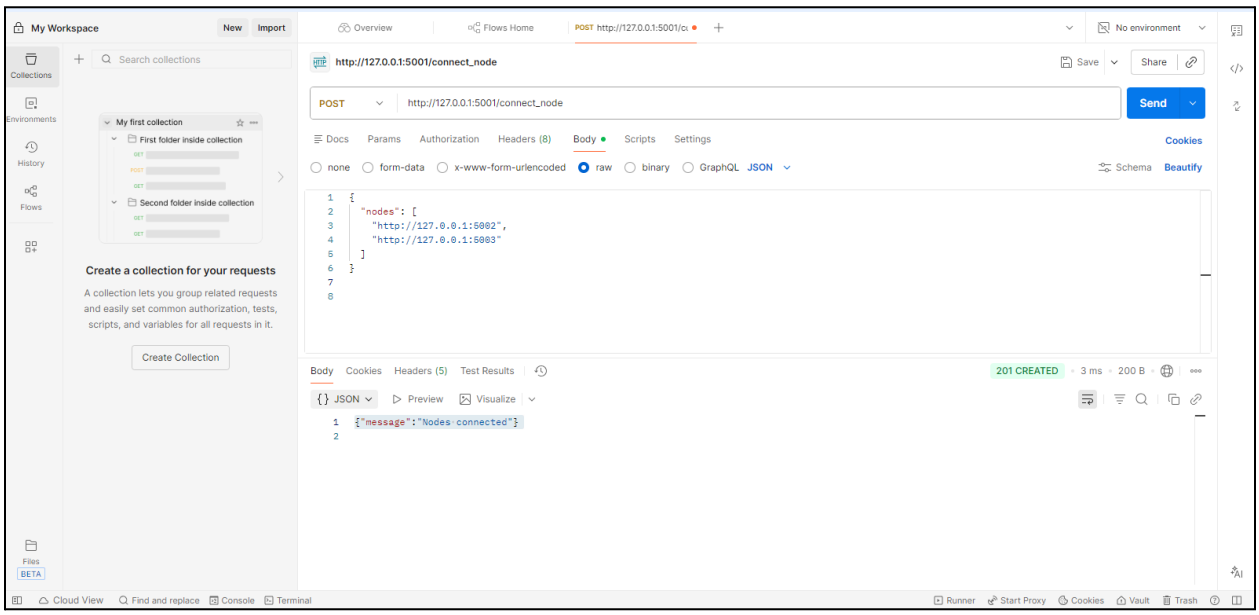
"transactions": [
{
"amount": 10,
"receiver": "b",
"sender": "a"
},
{
"amount": 1,
"receiver": "Sneha",
"sender": "666a1298e4b94260b790aa2cd1d0e889"
}
]
}
],
"length": 3
}
Files
BETA
Cloud View
Find and replace
Console
Terminal
Runner
Start Proxy
Cookies
Vault
Trash


My Workspace
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Environments
History
Flows
Search collections
My first collection
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Second folder inside collection
Create a collection for your requests
A collection lets you group related requests and easily set common authorization, tests, scripts, and variables for all requests in it.
Create Collection
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Send
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Body
Scripts
Settings
Cookies
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raw
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GraphQL
JSON
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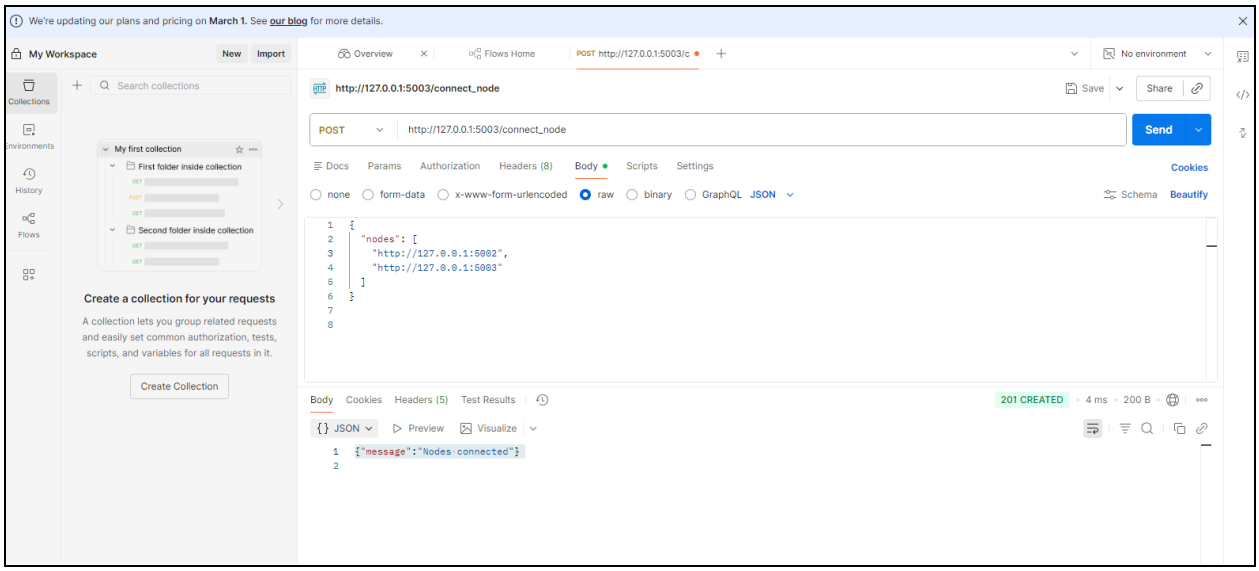
Beautify
{
"nodes": [
"http://127.0.0.1:5002",
"http://127.0.0.1:5003"
]
}
Body
Cookies
Headers (5)
Test Results
201 CREATED
3 ms
200 B
JSON
Preview
Visualize
{"message":"Nodes connected"}
Files
BETA
Cloud View
Find and replace
Console
Terminal
Runner
Start Proxy
Cookies
Vault
Trash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My Workspace
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Overview
Flows Home
POST http://127.0.0.1:5003/c
No environment
Collections
Environments
History
Flows
Search collections
My first collection
First folder inside collection
Second folder inside collection
Create a collection for your requests
A collection lets you group related requests and easily set common authorization, tests, scripts, and variables for all requests in it.
Create Collection
http://127.0.0.1:5003/connect_node
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GraphQL
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{
"nodes": [
"http://127.0.0.1:5002",
"http://127.0.0.1:5003"
]
}
Body
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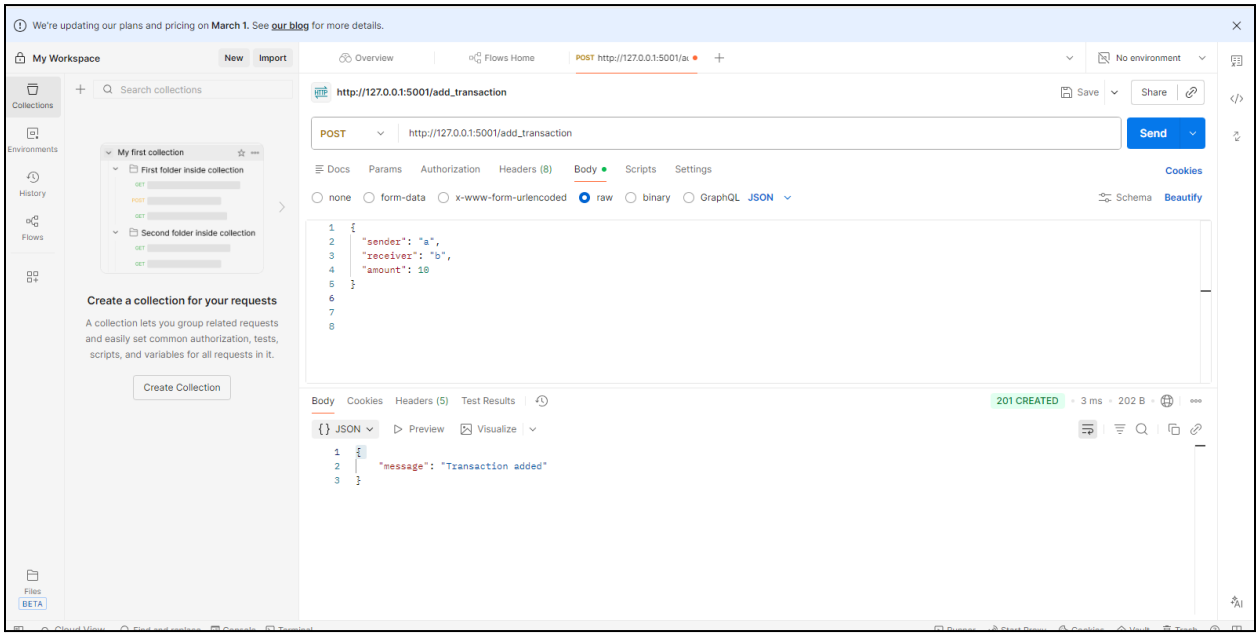
{"message":"Nodes connected"}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My Workspace
New
Import
Overview
Flows Home
POST http://127.0.0.1:5001/a
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Environments
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Flows
Search collections
My first collection
First folder inside collection
Second folder inside collection
Create a collection for your requests
A collection lets you group related requests and easily set common authorization, tests, scripts, and variables for all requests in it.
Create Collection
http://127.0.0.1:5001/add_transaction
Save
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POST
http://127.0.0.1:5001/add_transaction
Send
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Authorization
Headers (8)
Body
Scripts
Settings
Cookies
none
form-data
x-www-form-urlencoded
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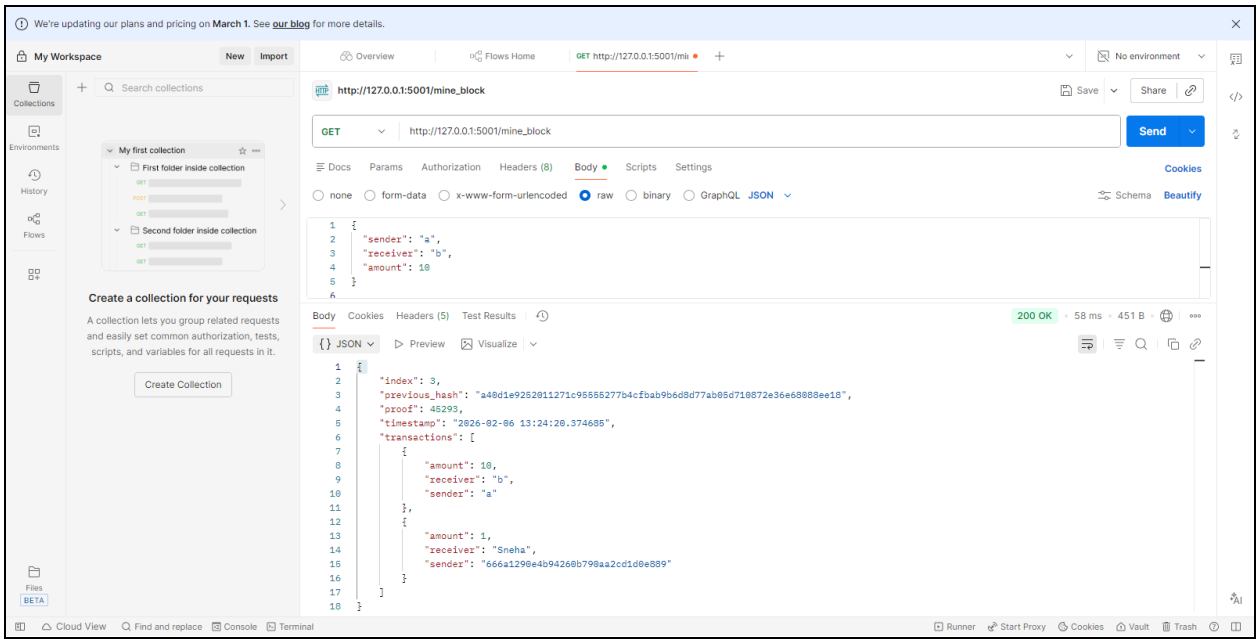
binary
GraphQL
JSON
Schema
Beautify
{
"sender": "a",
"receiver": "b",
"amount": 10
}
Body
Cookies
Headers (5)
Test Results
201 CREATED
3 ms
202 B
JSON
Preview
Visualize
{
"message": "Transaction added"
}
Files
BETA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My Workspace
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Search collections
My first collection
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Create a collection for your requests
A collection lets you group related requests and easily set common authorization, tests, scripts, and variables for all requests in it.
Create Collection
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http://127.0.0.1:5001/mine_block
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GraphQL
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Beautify
{
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"amount": 10
}
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Headers (5)
Test Results
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Preview
Visualize
{
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"transactions": [
{
"amount": 10,
"receiver": "b",
"sender": "a"
},
{
"amount": 1,
"receiver": "Sneha",
"sender": "666a1290e4b94260b798aa2cd1d0e889"
}
]
}
Cloud View
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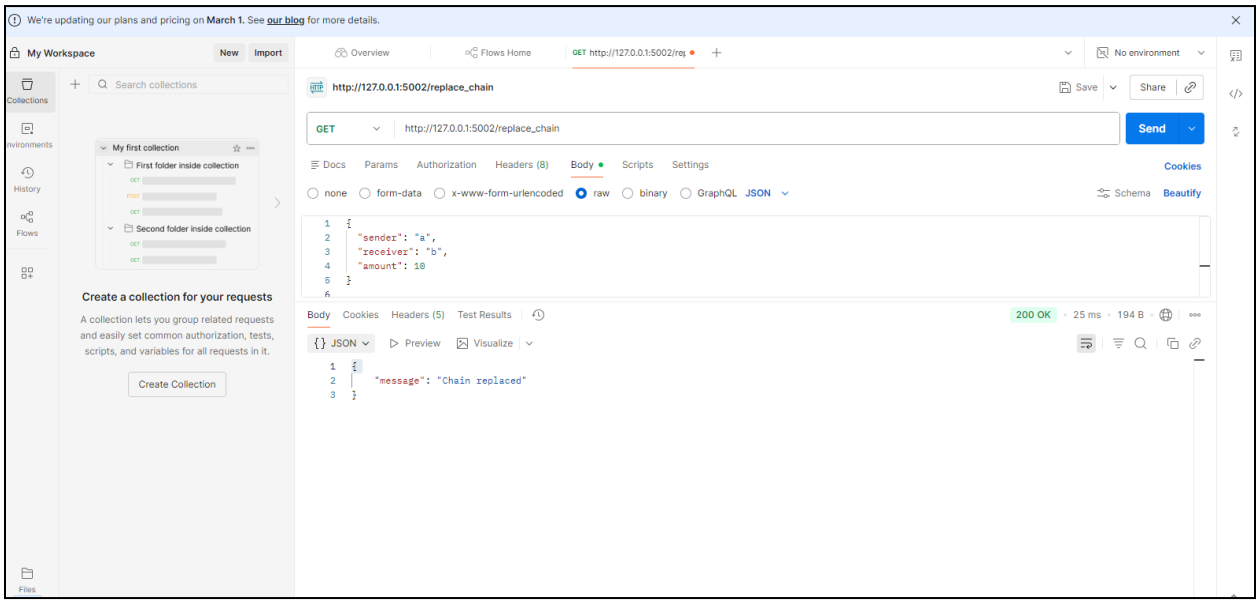
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My Workspace
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Second folder inside collection
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A collection lets you group related requests and easily set common authorization, tests, scripts, and variables for all requests in it.
Create Collection
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http://127.0.0.1:5002/replace_chain
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http://127.0.0.1:5002/replace_chain
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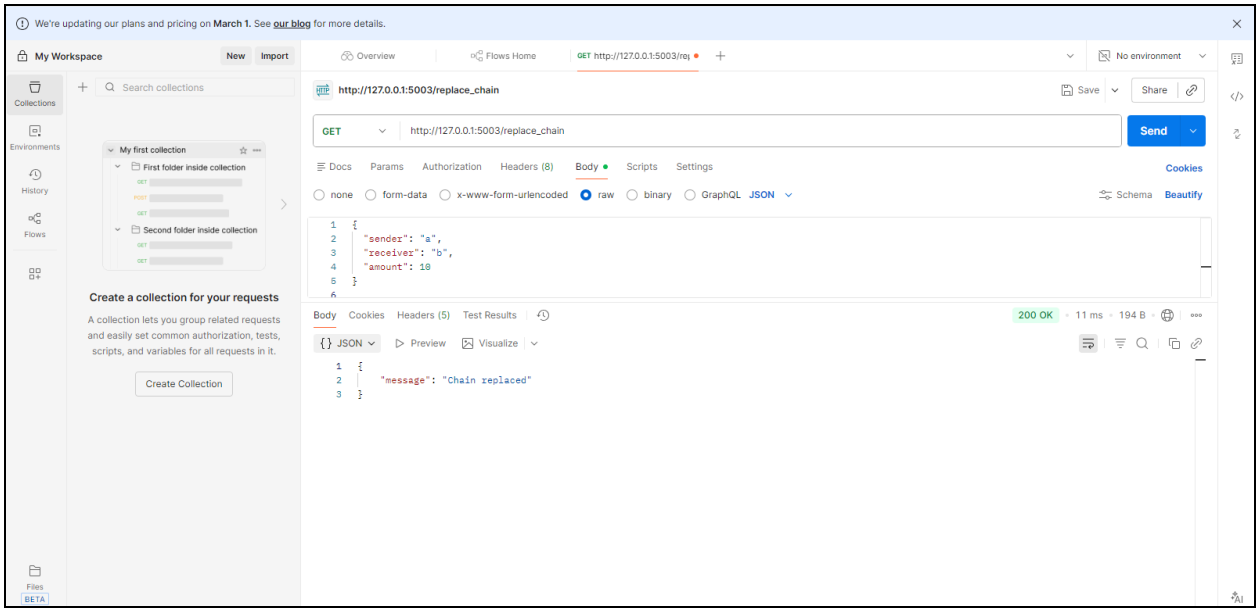
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Schema
Beautify
{
"sender": "a",
"receiver": "b",
"amount": 10
}
Body
Cookies
Headers (5)
Test Results
200 OK
25 ms
194 B
JSON
Preview
Visualize
{
"message": "Chain replaced"
}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raw
binary
GraphQL
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Schema
Beautify
{
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"amount": 10
}
Body
Cookies
Headers (5)
Test Results
200 OK
11 ms
194 B
JSON
Preview
Visualize
{
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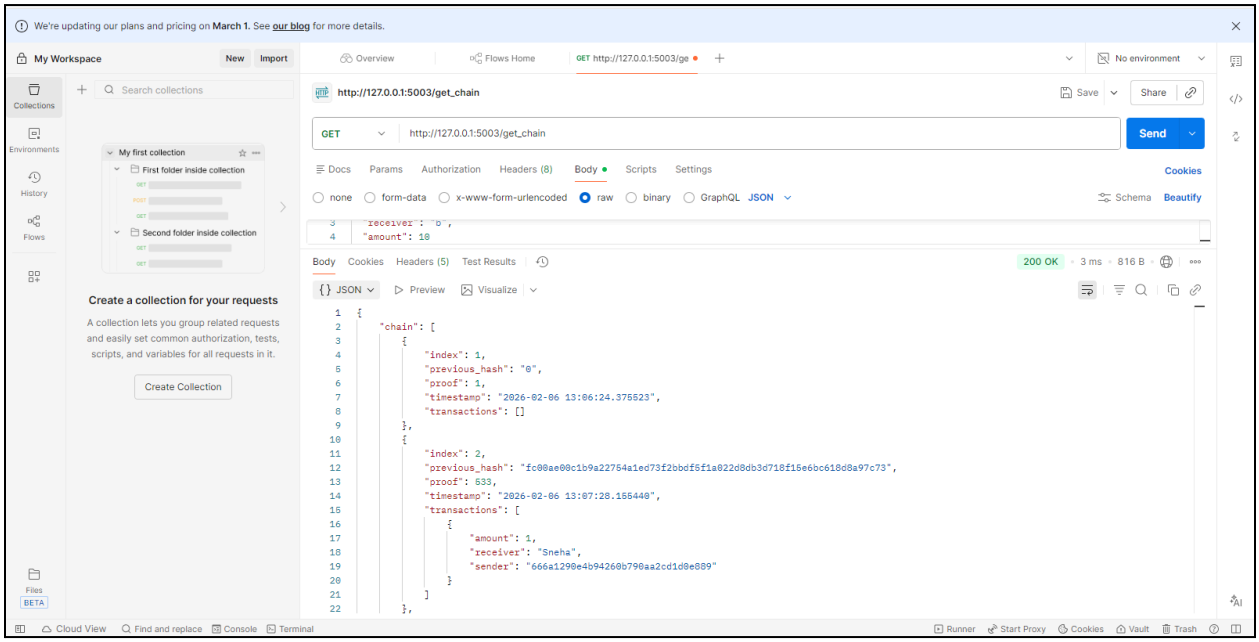
}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Headers (8)
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Scripts
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form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Schema
Beautify
"receiver": "b",
"amount": 10
Body
Cookies
Headers (5)
Test Results
200 OK
3 ms
816 B
JSON
Preview
Visualize
{
"chain": [
{
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"proof": 1,
"timestamp": "2026-02-06 13:06:24.375523",
"transactions": []
},
{
"index": 2,
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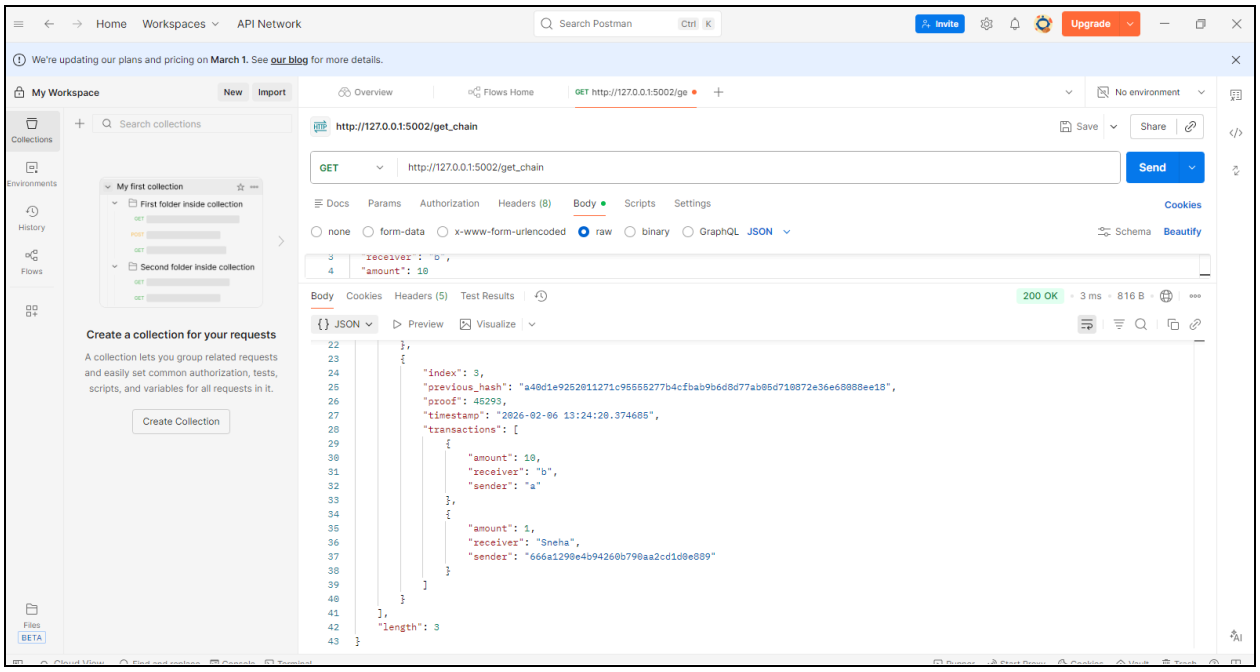
"proof": 633,
"timestamp": "2026-02-06 13:07:28.155440",
"transactions": [
{
"amount": 1,
"receiver": "Sneha",
"sender": "666a1290e4b94260b790aa2cd1d9e889"
}
]
},
Cloud View
Find and replace
Console
Terminal
Runner
Start Proxy
Cookies
Vault
Trash


Home
Workspaces
API Network
Search Postman
Invite
Upgrade
We're updating our plans and pricing on March 1. See our blog for more details.
My Workspace
New
Import
Overview
Flows Home
GET http://127.0.0.1:5002/ge
No environment
Collections
Environments
History
Flows
Search collections
My first collection
First folder inside collection
Second folder inside collection
Create a collection for your requests
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Create Collection
http://127.0.0.1:5002/get_chain
Save
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http://127.0.0.1:5002/get_chain
Send
Docs
Params
Authorization
Headers (8)
Body
Scripts
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Schema
Beautify
"receiver": "b",
"amount": 10
Body
Cookies
Headers (5)
Test Results
200 OK
3 ms
816 B
JSON
Preview
Visualize
},
{
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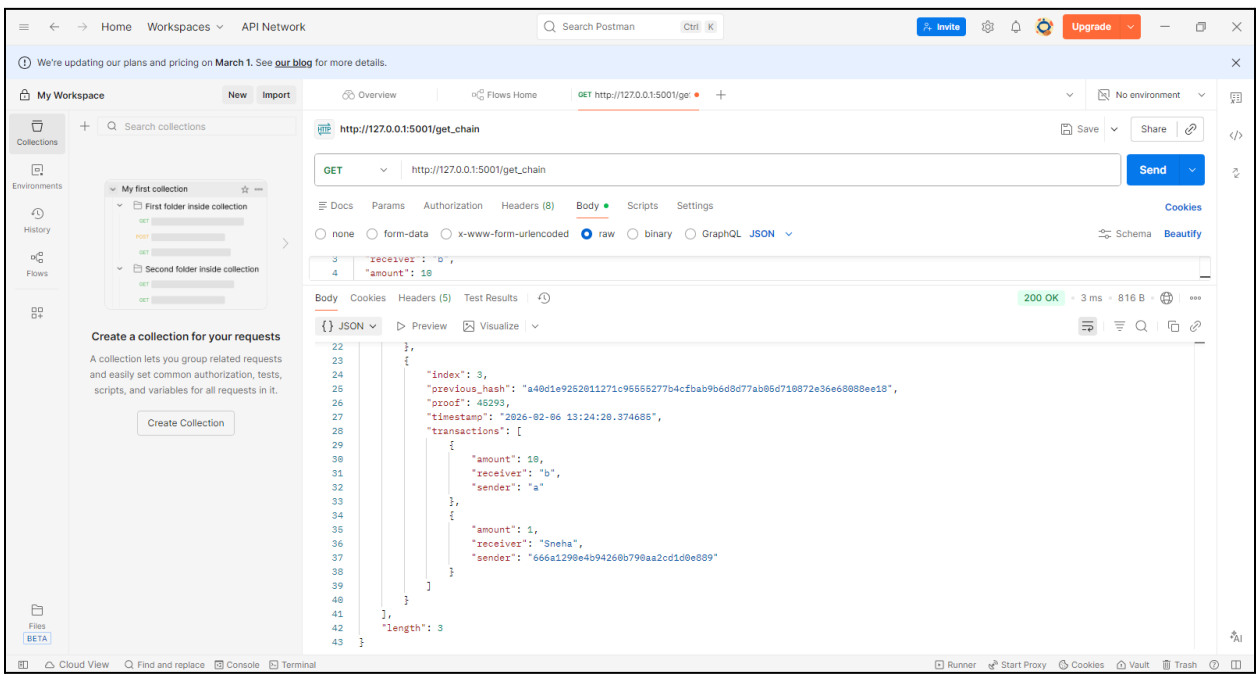
"proof": 45293,
"timestamp": "2026-02-06 13:24:28.374685",
"transactions": [
{
"amount": 10,
"receiver": "b",
"sender": "a"
},
{
"amount": 1,
"receiver": "Sneha",
"sender": "666a1290e4b94260b790aa2cd1d0e889"
}
]
}
],
"length": 3
}
Files
BETA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My Workspace
New
Import
Overview
Flows Home
GET http://127.0.0.1:5001/ge
No environment
Collections
Environments
History
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My first collection
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"amount": 10
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Cookies
Headers (5)
Test Results
200 OK
3 ms
816 B
JSON
Preview
Visualize
},
{
"index": 3,
"previous_hash": "a40d1e9252011271c95555277b4cfbab9b6d8d77ab05d710872e36e68088ee18",
"proof": 45293,
"timestamp": "2026-02-06 13:24:28.374685",
"transactions": [
{
"amount": 10,
"receiver": "b",
"sender": "a"
},
{
"amount": 1,
"receiver": "Sneha",
"sender": "666a1290e4b94260b790aa2cd1d0e889"
}
]
}
],
"length": 3
}
Files
BETA
Cloud View
Find and replace
Console
Terminal
Runner
Start Proxy
Cookies
Vault
Trash

http://127.0.0.1:5001/add_transaction

Save | Share

POST http://127.0.0.1:5001/add_transaction Send

Docs | Params | Authorization | Headers (8) | Body | Scripts | Settings | Cookies

none | form-data | x-www-form-urlencoded | raw | binary | GraphQL | JSON | Schema | Beautify

```
{
  "sender": "a",
  "receiver": "patra",
  "amount": 50
}
```

Home | Workspaces | API Network | Search Postman Ctrl K | Invite | Upgrade

We're updating our plans and pricing on March 1. See our blog for more details.

My Workspace | New | Import

Collections | Environments | History | Flows | Files BETA

Search collections

My first collection
- First folder inside collection
- Second folder inside collection

Create a collection for your requests

A collection lets you group related requests and easily set common authorization, tests, scripts, and variables for all requests in it.

Create Collection

Overview | Flows Home | GET http://127.0.0.1:5001/ge | No environment

http://127.0.0.1:5001/get_chain

Save | Share

GET http://127.0.0.1:5001/get_chain Send

Docs | Params | Authorization | Headers (8) | Body | Scripts | Settings | Cookies

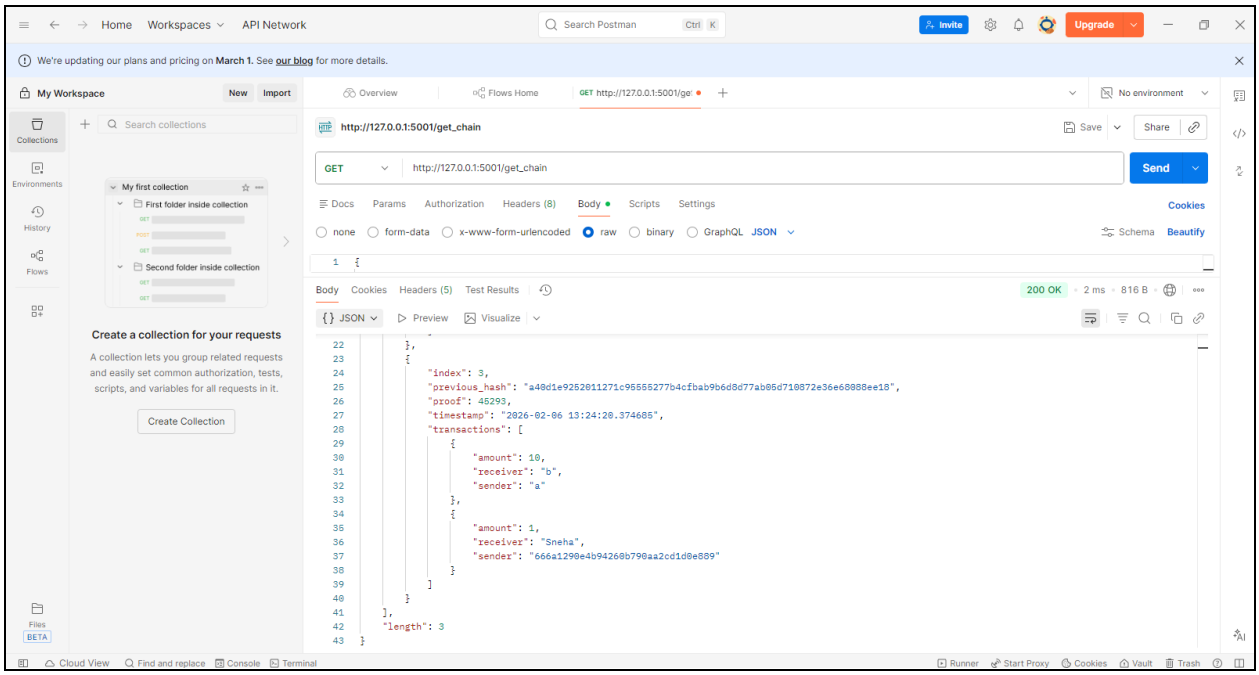
none | form-data | x-www-form-urlencoded | raw | binary | GraphQL | JSON | Schema | Beautify

Body | Cookies | Headers (5) | Test Results | 200 OK | 2 ms | 816 B

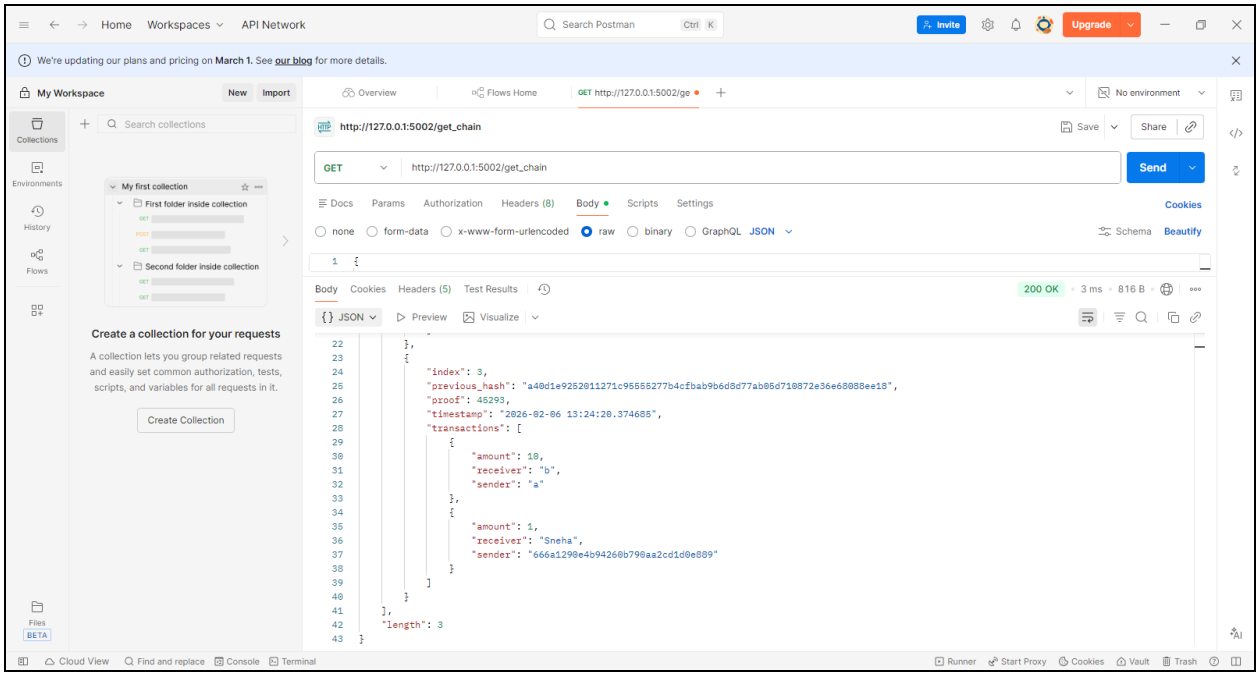
JSON | Preview | Visualize

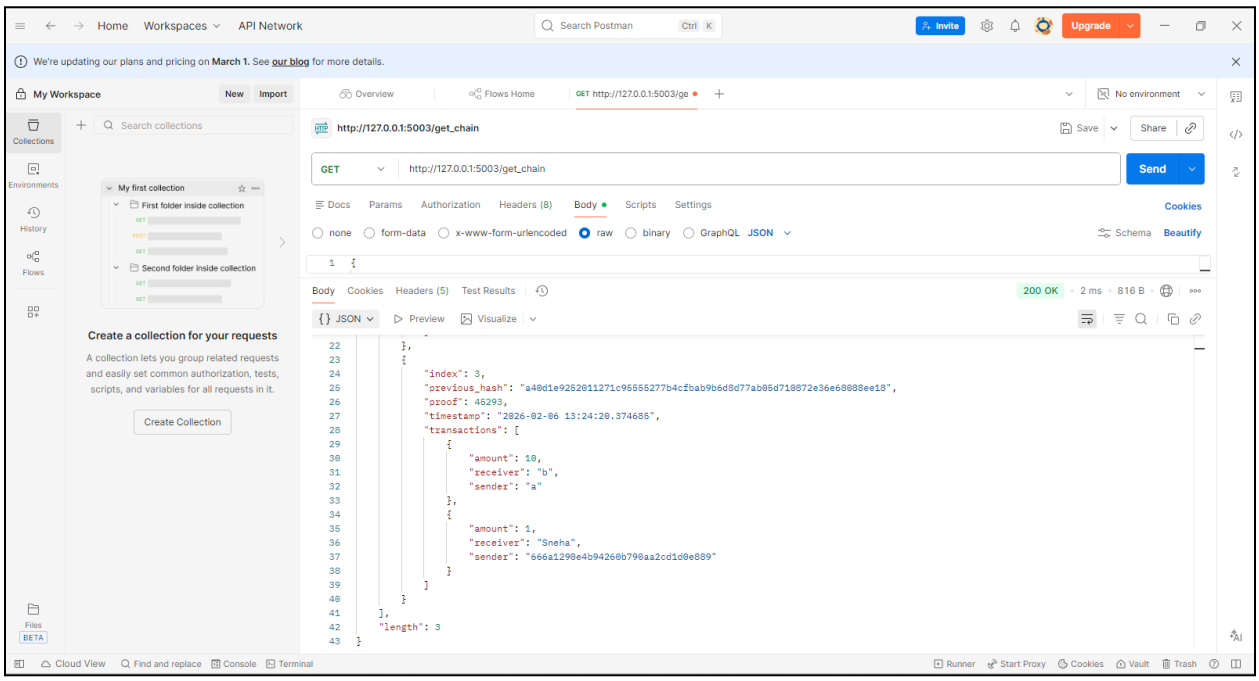

```
{
    "chain": [
        {
            "index": 1,
            "previous_hash": "0",
            "proof": 1,
            "timestamp": "2026-02-06 13:06:24.375523",
            "transactions": []
        },
        {
            "index": 2,
            "previous_hash": "fc00ae00c1b9a22754a1ed73f2bbdf5f1a022d8db3d718f15e6bc618d8a97c73",
            "proof": 533,
            "timestamp": "2026-02-06 13:07:28.155440",
            "transactions": [
                {
                    "amount": 1,
                    "receiver": "Sneha",
                    "sender": "666a1290e4b94260b790aa2cd1d0e889"
                }
            ]
        },
```


Cloud View | Find and replace | Console | Terminal | Runner | Start Proxy | Cookies | Vault | Trash




```
{
    "index": 3,
    "previous_hash": "a48d1e9252011271c95555277b4cfbab9b6d8d77ab05d710872e36e68988ee18",
    "proof": 45293,
    "timestamp": "2026-02-06 13:24:20.374685",
    "transactions": [
        {
            "amount": 10,
            "receiver": "b",
            "sender": "a"
        },
        {
            "amount": 1,
            "receiver": "Sneha",
            "sender": "666a1290e4b94268b790aa2cd1d8e889"
        }
    ]
}
],
"length": 3
}
```


Home
Workspaces
API Network
Search Postman
Invite
Upgrade
We're updating our plans and pricing on March 1. See our blog for more details.
My Workspace
New
Import
Overview
Flows Home
GET http://127.0.0.1:5002/ge
No environment
Collections
Environments
History
Flows
Search collections
My first collection
First folder inside collection
Second folder inside collection
Create a collection for your requests
A collection lets you group related requests and easily set common authorization, tests, scripts, and variables for all requests in it.
Create Collection
http://127.0.0.1:5002/get_chain
Save
Share
GET
http://127.0.0.1:5002/get_chain
Send
Docs
Params
Authorization
Headers (8)
Body
Scripts
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Schema
Beautify
Body
Cookies
Headers (5)
Test Results
200 OK
3 ms
816 B
JSON
Preview
Visualize
"index": 3,
"previous_hash": "a40d1e9252011271c95555277b4cfbab9b6d8d77ab05d710872e36e68088ee18",
"proof": 45293,
"timestamp": "2026-02-06 13:24:20.374685",
"transactions": [
"amount": 10,
"receiver": "b",
"sender": "a"
"amount": 1,
"receiver": "Sneha",
"sender": "666a1290e4b94260b790aa2cd1d0e889"
"length": 3
Cloud View
Find and replace
Console
Terminal
Runner
Start Proxy
Cookies
Vault
Trash


Home
Workspaces
API Network
Search Postman
Invite
Upgrade
We're updating our plans and pricing on March 1. See our blog for more details.
My Workspace
New
Import
Overview
Flows Home
GET http://127.0.0.1:5003/ge
No environment
Collections
Environments
History
Flows
Search collections
My first collection
First folder inside collection
Second folder inside collection
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Create Collection
http://127.0.0.1:5003/get_chain
Save
Share
GET
http://127.0.0.1:5003/get_chain
Send
Docs
Params
Authorization
Headers (8)
Body
Scripts
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Schema
Beautify
Body
Cookies
Headers (5)
Test Results
200 OK
2 ms
816 B
JSON
Preview
Visualize
"index": 3,
"previous_hash": "a40d1e9252011271c95555277b4cfbab9b6d8d77ab05d710872e36e68088ee18",
"proof": 45293,
"timestamp": "2026-02-06 13:24:20.374685",
"transactions": [
"amount": 10,
"receiver": "b",
"sender": "a"
"amount": 1,
"receiver": "Sneha",
"sender": "666a1290e4b94260b790aa2cd1d0e889"
"length": 3
Cloud View
Find and replace
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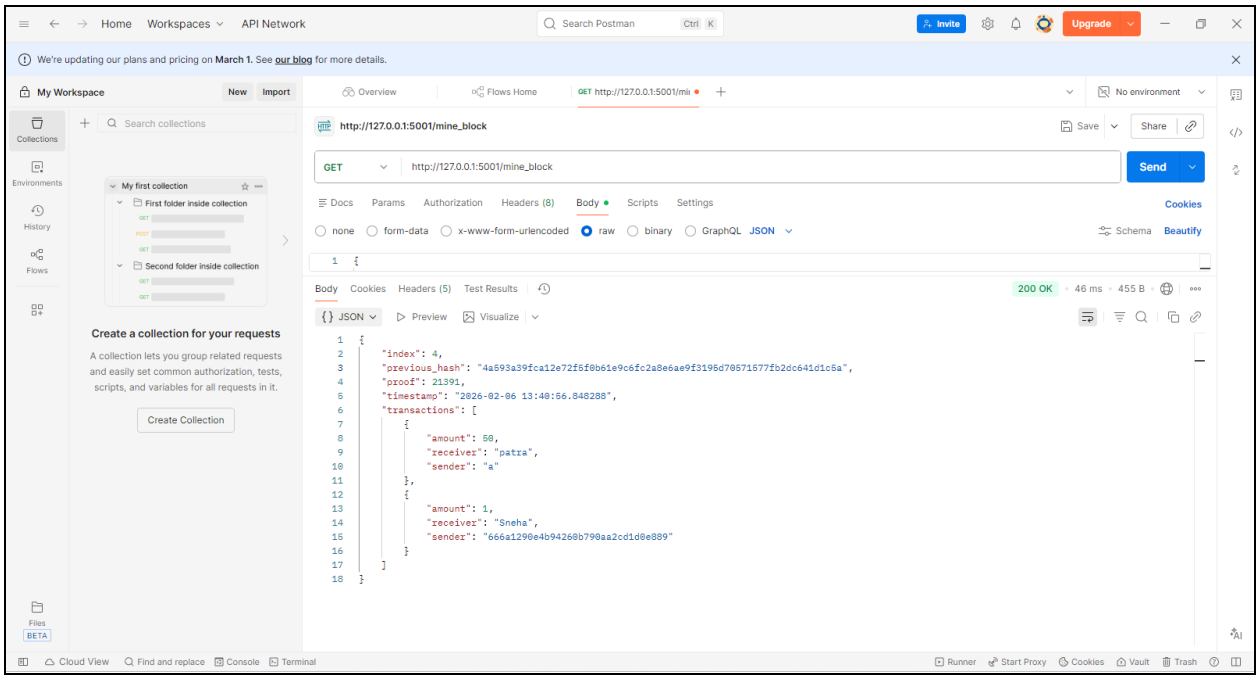
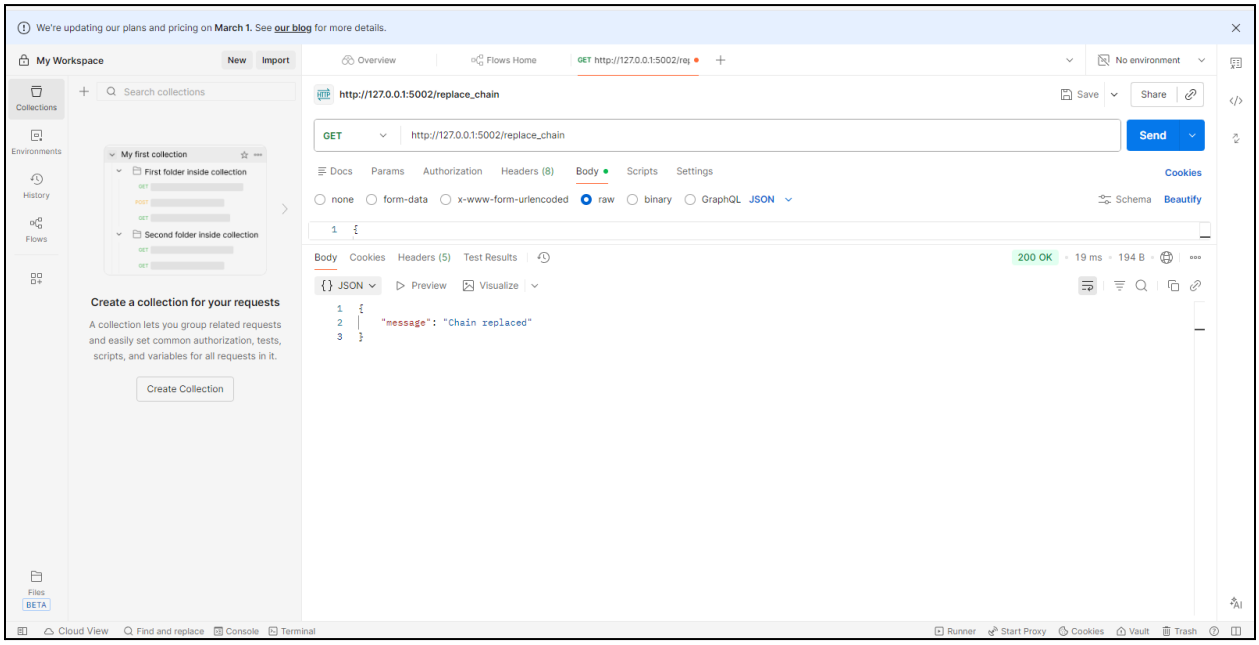
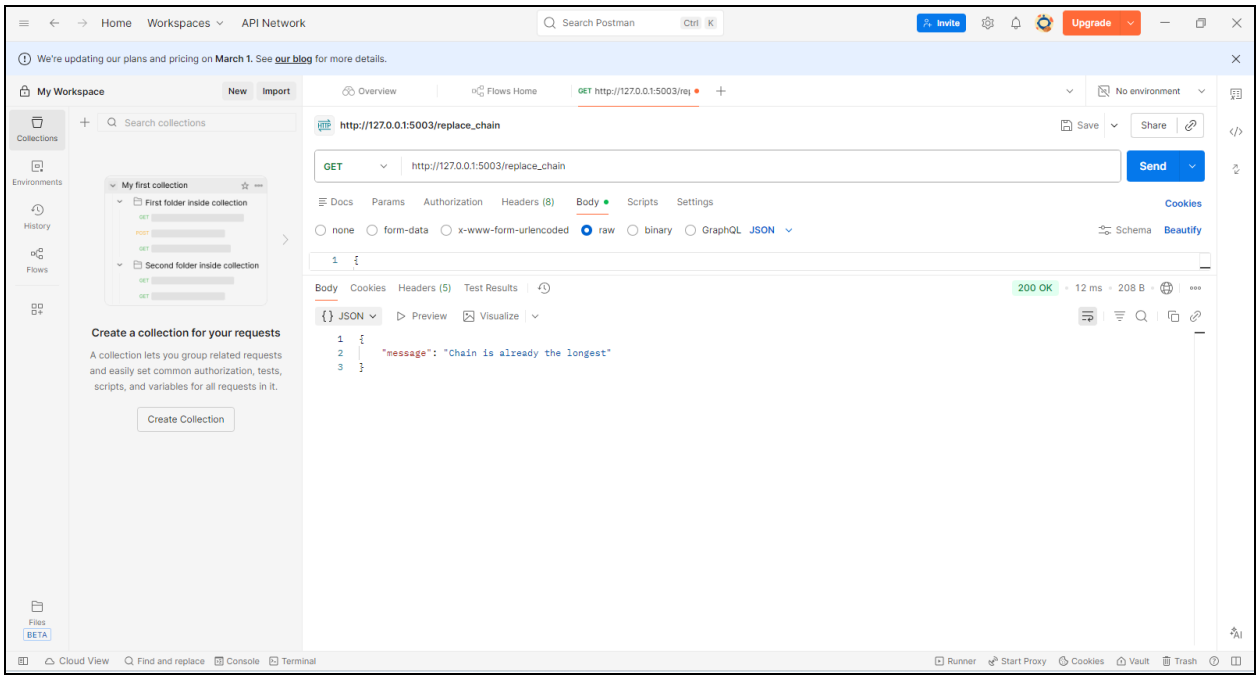
Trash


GET http://127.0.0.1:5001/mine_block
Send
Body
200 OK
46 ms
455 B
{
"index": 4,
"previous_hash": "4a593a39fca12e72f5f0b61e9c6fc2a8e6ae9f3195d70571577fb2dc641d1c5a",
"proof": 21391,
"timestamp": "2026-02-06 13:48:56.848288",
"transactions": [
{
"amount": 50,
"receiver": "patra",
"sender": "a"
},
{
"amount": 1,
"receiver": "Sneha",
"sender": "666a1290e4b94268b790aa2cd1d0e889"
}
]
}


GET http://127.0.0.1:5002/replace_chain
Send
Body
200 OK
19 ms
194 B
{
"message": "Chain replaced"
}



Conclusion:

We developed a cryptocurrency using Python and implemented mining in a simulated three-node blockchain network. Each node maintained its own ledger and synchronized with peers using the Longest Chain Rule to ensure consistency. Mining was performed through Proof-of-Work, securely adding transactions and rewarding miners. This demonstrated key blockchain concepts, including decentralized consensus, transaction validation, and network synchronization.